Outcomes of Multidisciplinary Low Vision Rehabilitation in Adults

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Purpose. To evaluate the outcomes of multidisciplinary low vision rehabilitation (LVR) in adults with low vision (LV) in India using the Veterans Affairs Low Vision Visual Functioning Questionnaire (VA LV VFQ-48) and the Impact of Vision Impairment (IVI) questionnaire.

Methods. Consecutive adults with LV referred for the first time to the Centre for Sight Enhancement were administered the 28-item IVI and VA LV VFQ-48 before and after LVR (4 months). Rasch-scaled scores were generated for the IVI and the VA LV VFQ-48 along with each of their subscales at both the time points. Effect size (ES) was calculated using Cohen's *d* coefficient.

Results. Two hundred fifty-five patients completed the LVR program. Following LVR, the linear measures revealed significant improvements in vision-related quality of life (VRQoL) using the IVI at follow-up ($P < 0.0001$), with large ES for reading and accessing information (ES = 1.0) and relatively moderate ES for overall IVI (ES = 0.63) and mobility subscale (ES = 0.53). No significant improvement was found on emotional well-being subscale (ES = 0.18; $P = 0.06$). Similarly, significant improvements in VRQoL ($P < 0.0001$) were found for the VA LV VFQ-48 at follow-up, albeit moderate ES for the overall visual ability (ES = −0.67) and its subscales (reading [ES = −0.72], mobility [ES = −0.45], visual information [ES = −0.47), visual motor [ES = −0.54]).

Conclusions. Regardless of the instrument used for assessment, our multidisciplinary LVR program resulted in statistically significant improvements in visual functioning and VRQoL of those adults with LV deemed appropriate for intervention (dropout rates were high); however, the improvements were modest, and the clinical relevance of these small improvements may be limited. Nonetheless, further studies should attempt to increase the completion rates in order to draw strong inferences.

Keywords: outcomes, low vision rehabilitation, visual impairment, Rasch analysis, effectiveness, low vision devices, impact of vision impairment, Veterans Affairs Low Vision Visual Functioning questionnaire

Low vision (LV) refers to a condition of reduced vision that is uncorrectable by standard glasses, contact lenses, medication, or surgery and that impacts a person's ability to perform age-appropriate vision-dependent tasks.[1] Recent published data suggest that, worldwide, there were 191 million people with LV in 2010.[2] However, up to 90% of patients with LV have useful residual vision and may benefit from low vision rehabilitation (LVR).[3–5] Low vision rehabilitation aims to improve functional ability and independence of the visually impaired, and as a result to improve their quality of life (QoL).[6] Therefore, LVR is considered a highly valued intervention that has a significant impact on individuals' lives and activities.[7] The reported success rates of LVR across studies have typically ranged from 23% to 100%.[8] In all these studies, the outcome measures for success have ranged from subjective ratings that involve asking patients about their satisfaction with the LVR to objective indicators that involve using performance-based indicators of functional ability (such as reading speed and duration).[9–15] However, a consistent definition of success is lacking.[7]

In the past decade there has been a paradigm shift in the evaluation of LVR services, and the move has been toward the use of patient-reported outcome measures (PROMs) that involve obtaining patients' perception of functional ability before and after LVR. Several researchers have used PROMs to evaluate the effectiveness of LVR in improving the visual ability as well as the vision-related quality of life (VRQoL).[3,4,6,16–20] For example, in the Veterans Affairs LVR programs (both inpatient and outpatient), Stelmack et al.[19] used the Veterans Affairs Low Vision Visual Function Questionnaire-48 (VA LV VFQ-48) and demonstrated a very large positive effect (effect size [ES] ranging from 1.2 for mobility to 3.5 for reading ability) on self-reported functional ability of adults with LV.

Other LVR programs have shown a medium effect on functional ability.[21] In a recent study, Massof and Stelmack[22] compared the results of several LVR outcome studies and groups within studies conducted over the last 15 years and found that ES for most studies were small (<0.3) or moderate (0.5), except for the Veterans Affairs Blind Rehabilitation Centre (2.1) and Low Vision Intervention Trial. Using the National Eye Institute-Visual Function Questionnaire (NEI-VFQ-39) in visually impaired veterans before and after 2 months of treatment in an inpatient blind rehabilitation program, Kuyk et

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al.[23] reported medium to large effects (ES ranging from 0.26 for color vision to 1.97 for near vision subscale). By comparison, Stelmack et al.[6] reported small effects (increase in visual ability of 0.5 and 0.35 logits) using the NEI-VFQ-25 from the inpatient Veterans Affairs LVR program. Using the NEI-VFQ-51 and the Visual Function-14, Scott et al.[24] reported medium effects (ES ranging from 0.32 for general vision to 0.57 for near activities) from their LVR programs in a LV clinic sample. Lamoureux et al.[4] reported small to medium effects (ES ranging from 0.17 for mobility and independence to 0.30 for emotional well-being [EWB]) using the Impact of Vision Impairment (IVI), in terms of the changes in VRQoL from pre-LVR to post-LVR in their population. Similarly, using the IVI and the VA LV VFQ-48, Wang et al.[5] reported small to medium effects (ES ranging from 0.10 for visual information using the VA LV VFQ-48 to 0.24 for overall visual ability using the IVI) at 3 months following LVR in their population. DeBoer et al.[25] demonstrated a small effect (ES ranging from 0.13 to 0.17 using the Vision Quality of Life Core Measure 1, VCM1) at 1 year following LVR in their sample consisting of patients who received standard optometric as well as those who received multidisciplinary LV services. In the study by Reeves et al.,[17] the authors aimed to detect a standardized difference of 0.46 in the VRQoL between the three groups of patients in the trial (conventional LVR [CLVR], enhanced LVR [ELVR], and controlled for additional contact time in enhanced LVR [CELVR]), but failed to detect this difference using the VCM1 between those patients who were provided CLVR versus those who were provided ELVR or those who were provided CELVR. With the anticipated increase in the demand for LVR services and limited health care resource allocation, it is imperative that service providers demonstrate the effectiveness of their services in order to ensure continued funding of their programs.[26]

Models of LVR services vary widely in content, intensity, and mode of delivery.[27] However, there is consensus among service providers that LVR services should include an evaluation of the needs and goals of patients with LV and subsequently tailor the service provision accordingly.[28] There is also wide variability in the availability of LVR services across countries, with the coverage adequate in developed compared to that in the developing countries.[29] In any case, the current global coverage and the availability of LVR services in most countries are <10%. Part of the reason for such low coverage has been the location of the LVR services that are mainly at the secondary and tertiary levels, especially in developing countries.[30]

The Vision Rehabilitation Centres (VRC) of the L V Prasad Eye Institute, LVPEI (a tertiary eye care center), Hyderabad, located in the Southern state of Telangana (35.2 million population in 2011), India, has been a recognized provider of comprehensive LVR services in the country since 1997. It has served over 50,000 patients with LV to date and is one of the three major hospital-based LVR service providers in South India for a population of 252 million (approximately 25% of India's population). Though these services have been provided for over a decade in the region, the effectiveness of these LVR services in India has remained largely unknown. Recently, Do et al.[31] published the results of a survey that used the Low Vision Quality of Life (LVQoL) questionnaire to evaluate the effectiveness of LVR services in improving the QoL of patients with LV in a rural population of the neighboring Indian state Tamil Nadu. The service provided was, however, primarily clinical LV care that involved prescription of low vision devices (LVDs). The authors reported significant gains in mobility, distance vision and lighting adjustment, and reading and fine work, but did not find any significant change in activities of daily living of their patients following LVR.[31] However, these results may not apply to other regions/centers that follow a different LVR program. In addition, the limited demonstrable benefits of LVR in a selected population of South India calls for further investigation. With all this taken together and the fact that India is among the regions with the highest burden of visual impairment (VI) with limited health care resources, it is important for more such studies to be conducted in order to further explore the effectiveness of multidisciplinary LVR services in the country. The aim of the present prospective observational study was to evaluate the outcomes of multidisciplinary LVR services in adults with LV in Southern India using two popular Rasch-validated VRQoL instruments: the IVI and the VA LV VFQ-48. Vision-related QoL is a complex trait that encompasses visual disability, symptoms, EWB, social relationships, concerns, and convenience as they are affected by vision.[32] We chose the IVI and VA LV VFQ-48 given that they been validated extensively using Rasch analysis and that a change in VRQoL has been demonstrated in three LV services by three independent research teams in other populations.[4–6] More importantly, they enable measurement of different aspects of VRQoL that are considered important by patients with LV.

Methods

We recruited consecutive adults referred to the Centre for Sight Enhancement (CSE), VRC, at the LVPEI, Hyderabad, India, for the management of LV between March 2012 and March 2014. All the adults were referred from the outpatient services and specialty clinics of the LVPEI. The initial detailed ophthalmologic examinations established the diagnosis and cause for LV in the clinics. Participants were those who were first-visit patients of CSE; had self-reported difficulty with performing everyday tasks; could perform standard clinical vision tests; could speak English, Telugu, or Hindi; and were able to respond to the items (questions) on the instruments (details later). We included adults with visual acuity of 20/20 in the better eye ($n = 3$) because they had accompanying visual field loss in this eye (less than 20°). We excluded participants with disabilities such as hearing and intellectual disabilities because of the difficulty in obtaining reliable responses from them during administration of the instruments. In addition, we excluded patients who were prior users of LVDs; were unable to come for a scheduled follow-up; were unable to speak English, Hindi or Telugu; or had bilateral no perception of light. Ethical approval was obtained from the Ethics Committee for Human Research of the LVPEI, and the research adhered to the tenets of the Declaration of Helsinki. Informed consent was obtained from the participants and their parent(s) or caregiver after a detailed explanation of the study.

Instruments

Impact of Vision Impairment (IVI). The IVI is a vision-specific QoL instrument that was developed to measure the impact of LVR on participation in daily living in patients with LV.[33–37] The original instrument consisted of 32 items, but it was shortened to 28 items through Rasch analysis few years ago.[35] The 28 items are divided into three subscales: mobility and independence (11 items), EWB (8 items), and reading and accessing information (9 items). The questionnaire responses were organized and assigned ordinal values as recommended by the developers. The questionnaire uses two different rating scales: one for the first 15 items and the second for the remaining 13 items. Using standard procedures, local language versions were obtained using forward–backward translations of the IVI questionnaire.

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Veterans Affairs Low Vision Visual Functioning Questionnaire-48 (VA LV VFQ-48). The VA LV VFQ-48 is a vision-specific instrument that was developed as an outcome measure for LVR.[38–40] The 48-item content reflects activities that are important to most patients, depend heavily on vision, and are targeted by LVR.[41,42] It was validated using Rasch analysis and has been shown to be sensitive to change as a result of LVR.[38,39] All the items utilize a five-category response format, and the method of administration followed was as recommended by Stelmack and Massof.[43]

While the IVI provides an assessment of EWB (albeit a restricted assessment of VRQoL), the VA LV VFQ-48 provides a comprehensive assessment of the visual disability. Taken together, the IVI and the VA LV VFQ-48 complement each other in the assessment of the VRQoL.

The Low Vision Rehabilitation Program at the LVPEI. Details of the comprehensive LVR program at VRC, LVPEI have been published earlier.[44] We provide only the relevant details of the program here. Two vision rehabilitation centers (located next to each other) of the LVPEI provided the comprehensive LVR service, namely, the Meera and L B Deshpande Centre for Sight Enhancement and the Dr P R K Prasad Centre for Rehabilitation of the Blind and Visually Impaired. The multidisciplinary team comprised several professionals that included optometrists trained specifically in LV, special educators, orientation and mobility (O and M) instructors, rehabilitation professionals, information and communication technology instructors, and physiotherapists. The LVR commenced with the optometrist's performing the clinical assessments, and LVDs were demonstrated depending on the requirement of the task, as well as magnification assessment for distance and near. The optical LVDs included telescopes and magnifiers (stand and pocket). Nonoptical devices included reading stand, reading lamp, filter for glare control, and needle threader. Electronic devices included portable video magnifiers and desktop closed-circuit television (CCTV). Following the choice of the LVDs, patients were provided training in use of the prescribed LVDs. In addition, training in O and M, in use of computers (special software: screen magnification and screen reading), and in activities of daily living, audio books and disability certification were among the services provided by the rehabilitation professionals at the center. Finally, the LVDs were dispensed at the center and a review appointment was made within 4 months (±2 weeks) to assess the use of LVDs by the patient and to evaluate the effectiveness of the intervention. Two or three telephone reminders (separated by a week) were provided for those who failed to keep their follow-up appointment. If the patient failed to return for the follow-up visit, he or she was excluded from the study. Any LVD or glasses prescribed were to be purchased by the patient. Some patients who were economically underprivileged were provided waivers through an indigent fund available for this purpose.

Procedure

We administered both the questionnaires in English, Telugu, or Hindi, and this was done by trained interviewers in a face-to-face interview. We did not follow any specific order of administration of the questionnaires; they were administered at baseline and subsequently at the follow-up visit scheduled 4 months from the baseline. This duration was considered to be adequate for patients to have gained some experience in the use of the prescribed LVDs. The interview was conducted using a written script by a different trained interviewer at each visit. These interviewers were not involved in the LVR process and were masked to the results of the baseline data of the questionnaires. Demographic and clinical data were abstracted from the medical records of the patients.

Statistical Analysis

We used the formula presented by Lamoureux et al.[35] to transform the raw scores to Rasch-scaled scores for the IVI. Similarly we used the formula reported by Stelmack and Massof[43] to transform the raw scores to Rasch-scaled scores for the VA LV VFQ-48. While higher positive scores represent greater visual disability for the VA LV VFQ-48, they represent better VRQoL for the IVI.

Given that we used the Rasch-scaled scores, the instrument scores (total and subscale) for both the IVI and the VA LV VFQ-48 yielded continuous scores. We based our sample size calculation on the ES reported from similar studies in the literature (as noted in the introduction) and used a modest ES of 0.2 for our study. This calculation provided a sample size of 196 participants to detect an ES of 0.20 with 80% power at a 5% (2-tailed) significance level—that is, a "small ES." However, our clinic experience suggested a high attrition rate (almost 80%), and based on recommended calculations,[45] the minimum required sample size was 980 (accounting for 80% attrition rate) in order to ensure that the required final sample size of 196 was achieved in our study.

As noted in the introduction, we used the outcome measures from both these instruments so as to enable comparison of our results with previous studies and to be as comprehensive in our assessments of VRQoL as possible. We used SPSS statistical software to perform descriptive statistics (Version 19.0; SPSS Science, Chicago, IL, USA). We used χ^2 tests for categorical variables and independent samples *t*-tests for continuous variables as appropriate. We used a two-sided paired *t*-test at 5% significance level to assess whether the change in visual functioning (VF) and VRQoL scores from baseline to follow-up for both the instruments was significantly differently from zero. We employed mixed between–within subjects analysis of variance (ANOVA) to determine whether the LVR change was the same over time across participants' age (split at median; ≤40 vs. >40 years), sex, cause of VI (retinal versus nonretinal), education status, occupation status, presence or absence of systemic comorbidity, LV (best-corrected acuity of <20/60–20/200 in better eye) or legally blind (best-corrected acuity of <20/200 or visual field of <10° in better eye), location of residence (urban versus rural), and components of the multidisciplinary LVR service accessed by the participants (optometry led versus other services). We used the Bonferroni adjustment for multiple comparisons when appropriate. We calculated the LVR outcomes using ES (i.e., Cohen's *d*)[46] and used the MedCalc for Windows, version 14.8 (MedCalc Software, Ostend, Belgium) for this purpose. Effect size is defined as mean instrument score change divided by the standard deviation (SD) of the instrument scores at baseline. We used the conventional benchmarks to classify ES: small (≤0.2), moderate (0.3–0.7), and large (≥0.8).[46,47] In the presence of incomplete data it is imperative to perform sensitivity analyses, which examine the robustness of the results to the assumptions made in the primary analysis.[48,49] Therefore we performed sensitivity analyses using the baseline-observation-carried-forward (BOCF) method to handle participant drop-outs (see Results section).[50] With this method, the baseline observation is treated as the final response from the participant without any postbaseline observation regardless of the reason the participant dropped out or the score at the time of withdrawal.[50]

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Table 1. Reasons for Exclusion From the Study ($n = 2525$)

Reason	*n* (%)
Current/prior user of LVDs	20 (0.8)
Inability to return for a follow-up	1275 (50.4)
Refusal because of lack of time for participation	650 (25.7)
Presence of additional disabilities: hearing, intellectual	55 (21.8)
Age < 18 y	200 (7.9)
Inability to speak/understand one of the three languages required for administration of the instruments: English, Hindi, Telugu	325 (12.9)

Results

Between March 2012 and March 2014, 3796 first-visit adults with LV were provided LVR services. Of these, 1271 participants (33.5%) were identified and recruited for this study. The reasons for exclusion of potential participants from the study ($n = 2525$) are provided in Table 1. As is evident from the table, the most common reason for exclusion was an inability to return for a follow-up (50.4%). Of the 1271 persons recruited, 255 (20%) completed follow-up assessments, a rate much lower than that reported for Western populations.[3–5,51] Common reasons for failure to return for a follow-up assessment (withdrawal) were lack of specific LVR goals, perceived cosmetic blemish from the use of LVDs, the need to travel long distances that imposed a financial burden on the family, other commitments, LVDs being unaffordable, refusal to accept vision loss, and inability to be contacted by phone/mail.

The sociodemographic and clinical characteristics of the 1271 participants who were recruited for the study are outlined in Table 2. This sample is representative of the adults referred for LVR services to VRC. As noted earlier, 255 of the 1271 (20%) completed the study (completed group) and the remaining (1016) participants (withdrawal group) failed to complete the study. While the two groups of participants were comparable across most of the baseline characteristics, there was a statistically significant, but weak, association between the rate of completion of the study, the location of residence ($\chi^2 = 8.81$, $df = 1$, $P = 0.003$; $\phi = 0.09$), and the cause of VI ($\chi^2 = 13.27$, $df = 1$, $P = 0.02$; $\phi = 0.11$). A significantly higher proportion of urban participants (818, 80%), those with retinal disorders (556, 55%), and those without systemic comorbidity (711, 70%) withdrew from the study as compared to those from rural locations (198, 20%, $P = 0.002$), those without retinal disorders (460, 45%, $P = 0.005$), and those with systemic comorbidity (234, 23%, $P = 0.008$), respectively. When we compared the sociodemographics of those who were recruited for the study to those who were excluded, we found statistically significant differences between the groups (Table 3). Participant exclusion was significantly related to older age ($P < 0.0001$), being female ($P < 0.0001$), belonging to rural locations ($P < 0.0001$), being illiterate ($P = 0.01$), tendency to have retinal disorders as the cause of VI ($P = 0.05$), and having worse best-corrected visual acuity in the better eye ($P = 0.0003$) as compared to those who completed the study. The types of low vision interventions and various rehabilitation services undertaken by our participants are provided in Tables 4 and 5, respectively. Disability certification and, next, educational counseling were among the most frequently provided services by the rehabilitation professionals. The number of participants who purchased the LVDs is provided in the Figure.

Effectiveness of the Low Vision Rehabilitation Service Using the IVI

The overall IVI post-LVR score was significantly higher than the pre-LVR score ($P < 0.0001$) with ES of 0.63 (95% confidence interval [CI], 0.46–0.80) across all the participants (Table 6). Similarly, for the reading and accessing information subscale, the post-LVR score was significantly higher than the pre-LVR score ($P < 0.0001$) with ES of 1.00 (95% CI, 0.80–1.20). For the mobility and independence subscale, the post-LVR score was significantly higher than the pre-LVR score ($P < 0.0001$) with ES of 0.53 (95% CI, 0.31–0.75). By comparison, there was no significant difference in the EWB score between pre- and post-LVR ($P = 0.06$) (Table 6).

A mixed between–within subjects ANOVA conducted to compare the overall IVI and the two subscale scores between LV ($n = 214$) and legally blind ($n = 41$) groups across the two time points (baseline and follow-up) showed that there was no significant interaction between level of VI and the change in VRQoL for any of them (Wilks' lambda = 0.99, F (1, 253) = 1.28, $P = 0.28$ for overall IVI; Wilks' lambda = 1.00, F (1, 253) = 0.09, $P = 0.75$ for reading and accessing information subscale; Wilks' lambda = 0.99, F (1, 253) = 0.64, $P = 0.42$ for mobility and independence subscale). However, our study perhaps lacked sufficient power to detect the significant interaction effects (>0.5). There was a main effect for LVR (Wilks' lambda = 0.85, F (1, 253) = 44.9, $P < 0.0001$ for overall IVI; Wilks' lambda = 0.69, F (1, 253) = 113.38, $P < 0.0001$ for reading and accessing information subscale; Wilks' lambda = 0.86, F (1, 253) = 39.78, $P < 0.0001$ for the mobility and independence subscale) with the overall IVI and both the subscale scores improving significantly following LVR services across both groups of participants.

Effectiveness of the Low Vision Rehabilitation Service Using the VA LV-VFQ-48

For the VA LV VFQ-48, the overall post-LVR score was significantly higher than the pre-LVR score ($P < 0.0001$) with ES of –0.67 (95% CI, –0.46 to –0.89) across all the participants (Table 7). Similarly, for all four subscales, the post-LVR score was significantly higher than the pre-LVR score ($P < 0.0001$), and the ES was –0.72 (95% CI, –0.47 to –0.97), –0.45 (95% CI, –0.15 to –0.75), –0.54 (95% CI, –0.25 to –0.83), and –0.47 (95% CI, –0.20 to –0.74) for the reading, mobility, visual information, and visual motor subscales, respectively (Table 7).

A mixed between–within subjects ANOVA conducted to compare the overall VA LV VFQ-48 and the four subscale scores between LV and legally blind groups across the two time points showed that there was a significant interaction between level of VI and the change in visual disability for the overall VA LV VFQ-48 score (Wilks' lambda = 0.98, F (1, 253) = 4.80, $P = 0.03$). However, there was no significant interaction between level of VI and the change in visual disability for any of its subscales (Wilks' lambda = 0.99, F (1, 253) = 0.20, $P = 0.27$ for reading subscale; Wilks' lambda = 0.99, F (1, 253) = 2.03, $P = 0.16$ for mobility subscale; Wilks' lambda = 0.99, F(1, 253) = 1.86, $P = 0.17$ for visual information subscale; Wilks' lambda = 0.99, F (1, 253) = 3.24, $P = 0.07$ for visual motor subscale). However, our study perhaps lacked sufficient power to detect the significant interaction effects (>0.5). There was a main effect for LVR (Wilks' lambda = 0.83, F (1, 253) = 50.48, $P < 0.0001$ for overall VA LV VFQ-48; Wilks' lambda = 0.78, F (1, 253) = 69.39, $P < 0.0001$ for reading subscale; Wilks' lambda = 0.89, F (1, 253) = 30.55, $P < 0.0001$ for the mobility subscale; Wilks' lambda = 0.87, F (1, 253) = 38.61, $P < 0.0001$ for visual information subscale; Wilks' lambda = 0.90, F (1, 253) = 27.21, $P < 0.0001$ for visual

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Table 2. Sociodemographic and Clinical Characteristics of the Participants Who Completed This Study (Completed) and Those Who Failed to Complete the Study (Withdrawal)

Characteristic	Completed, $N = 255$ *n* (%) or Mean ± SD	Withdrawal, $N = 1016$ *n* (%) or Mean ± SD	*P*
Age, y	41.7 ± 16.1	41.3 ± 16.0*	0.72
Range	18–82	18–82	
Sex			
Male	196 (77)	764 (75)	0.64
Female	59 (23)	252 (25)	
Duration of VI, y†	10.4 ± 12.9	11.8 ± 13.3‡	0.13
Location of residence§			
Urban	183 (72)	818 (80)	0.003
Rural	72 (28)	198 (20)	
Educational status			
Literate	221 (87)	792 (78)	0.77
Illiterate	34 (13)	112 (11)	
Employment status			
Employed	137 (54)	528 (53)	0.78
Unemployed	118 (46)	478 (47)	
Systemic comorbidity‖			
Present	85 (33)	234 (23)	0.008
Absent	170 (67)	711 (70)	
Cause of VI			
Retinal disorders¶	114 (45)	556 (55)	0.005
Nonretinal disorders	141 (55)	460 (45)	
BCVA in better eye, Snellen	0.69 ± 0.41, 20/100	0.70 ± 0.40, 20/100#	0.72
Range, Snellen	0.0–2.4, 20/20–20/5000	0.0–2.4, 20/20–20/5000	
VI level, BCVA in better eye**			
≥20/60	82 (32)	319 (31)	0.29
<20/60–20/200	132 (52)	489 (48)	
<20/200	41 (16)	208 (20)	

BCVA, best-corrected visual acuity.

* Mean difference = –0.4 (95% CI, –2.6, 1.8).

† Time since diagnosis of eye condition.

‡ Mean difference = 1.4 (95% CI, –0.42, 3.22).

§ Self-reported.

‖ Includes diabetes mellitus, hypertension, coronary artery disease, asthma, thyroid disorders.

¶ Includes retinitis pigmentosa, cone–rod dystrophy, cone dystrophy, retinoschisis, diabetic retinopathy, age-related macular degeneration, familial exudative vitreoretinopathy, and so on.

Mean difference = 0.01 (95% CI, –0.04, 0.06).

** ≥20/60 includes those with visual field loss too and represents mild vision impairment; <20/60 to 20/200 represents moderate vision impairment; <20/200 represents severe vision impairment.

motor subscale), with the overall IVI and all four subscale scores improving significantly following LVR services across both the groups of participants.

Effectiveness of the Low Vision Rehabilitation at the Subgroup Level

There was no significant interaction effect between the LVR change and age (split at median: <40 vs. ≥40 years), sex, cause of VI (retinal versus nonretinal), education status (illiterate versus literate), occupation status (employed versus unemployed), presence or absence of systemic comorbidity, location of residence (urban versus rural), and type of service (optometry led versus other services). We also examined the effect of duration of vision loss and rehabilitative strategies provided by the LVR practitioner. Based on the median duration of vision loss (4 years), we divided our participants into two groups: less than median duration of vision loss and greater than median duration of vision loss. We found a significant interaction effect with LVR change (Wilks' lambda = 0.98, $F(1, 253) = 4.33$, $P = 0.04$) between these two groups for the overall IVI scores. There was a significant main effect for time, with both the groups showing an increase in the overall IVI scores at the follow-up visit (Wilks' lambda = 0.75, $F(1, 252) = 83.27$, $P < 0.0001$). Similarly, we found a significant interaction effect between the duration of vision loss and LVR change for the EWB subscale of the IVI (Wilks' lambda = 0.98, $F(1, 252) = 4.28$, $P = 0.04$) and for the mobility subscale of the VA LV VFQ-48 (Wilks' lambda = 0.98, $F(1, 252) = 6.09$, $P = 0.014$). There was a significant main effect for time, with both the groups showing an increase in the EWB subscale scores of the IVI (Wilks' lambda = 0.97, $F(1, 252) = 7.77$, $P = 0.006$) and the mobility scores of the VA LV VFQ-48 (Wilks' lambda = 0.80, $F(1, 252) = 61.70$, $P < 0.0001$) at the follow-up visit. Participants who had smaller duration

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Table 3. Comparison of Sociodemographic and Clinical Characteristics of the Participants Who Completed This Study With Those Who Were Excluded From the Study

Characteristic	Recruited, $N = 1271$ n (%) or Mean ± SD	Excluded, $N = 2525$ n (%) or Mean ± SD	P
Age, y	41.7 ± 16.1	47.4 ± 18.3*	<0.0001
Range	18–82	19–98	
Sex			
Male	196 (77)	1213 (48)	<0.0001
Female	59 (23)	1312 (52)	
Duration of VI, y†	10.4 ± 12.9	9.2 ± 12.0‡	0.15
Location of residence§			
Urban	183 (72)	1168 (46)	<0.0001
Rural	72 (28)	1357 (54)	
Educational status			
Literate	221 (87)	2019 (80)	0.01
Illiterate	34 (13)	506 (20)	
Employment status			
Employed	137 (54)	1295 (51)	0.47
Unemployed	118 (46)	1230 (49)	
Systemic comorbidity‖			
Present	85 (33)	809 (32)	0.67
Absent	170 (67)	1716 (68)	
Cause of VI			
Retinal disorders¶	114 (45)	1294 (51)	0.05
Nonretinal disorders	141 (55)	1231 (49)	
BCVA in better eye, Snellen	0.70 ± 0.4, 20/100	0.79 ± 0.5, 20/126^{+1}#	0.003
Range, Snellen	0.0–2.4, 20/20–20/5000	0.0–2.4, 20/20–20/5000	
VI level, BCVA in better eye**			
≥20/60	82 (32.2)	867 (34.3)	<0.0001
<20/60–20/200	132 (51.8)	992 (39.3)	
<20/200	41 (16.1)	666 (26.4)	

BCVA, best-corrected visual acuity.
* Mean difference = 5.7 (95% CI, 3.4, 8.0).
† Time since diagnosis of eye condition.
‡ Mean difference = −1.2 (95% CI, −2.8, 0.4).
§ Self-reported.
‖ Includes diabetes mellitus, hypertension, coronary artery disease, asthma, thyroid disorders.
¶ Includes retinitis pigmentosa, cone–rod dystrophy, cone dystrophy, retinoschisis, diabetic retinopathy, age-related macular degeneration, familial exudative vitreoretinopathy, and so on.
Mean difference = 0.09 (95% CI, 0.03, 0.16).
** ≥20/60 includes those with visual field loss too and represents mild vision impairment; <20/60 to 20/200 represents moderate vision impairment; <20/200 represents severe vision impairment.

of vision loss (≤4 years) exhibited significantly greater improvements in their EWB scores at the follow-up than those with greater duration of vision loss (>4 years) (1.07 ± 2.68 vs. 0.29 ± 2.80 logits, mean difference = 0.78 [95% CI, 0.10, 1.46], $P = 0.02$). We did not find any significant interaction effect for the other subscales when comparing participants with smaller versus greater duration of vision loss.

Sensitivity Analyses

The analyses were performed by carrying forward the baseline responses (i.e., the pre LVR) to the post-LVR data sheet for the withdrawal group, and we found a significant lowering of the ES using both the instruments. The ES ranged from 0.04 for EWB to 0.21 for reading using the IVI (Table 8) and from 0.07 for reading to 0.21 for overall visual ability using the VA LV VFQ-48 (Table 9).

Discussion

This study provides the first evidence that LVR at the VRC, LVPEI in South India, as measured by the two instruments—the IVI and the VA LV VFQ-48—resulted in significant improvements in the overall VF and the VRQoL of adults with LV in those who completed the study. Our results are in accordance with previous studies that have used these instruments and similar methodology to assess the impact of LVR services on the VRQoL in Western populations.[4,5,19,40]

In the only other recent study from India, Do et al.[31] administered the 25-item LVQoL questionnaire before and 1 month post LVR (over telephone) to participants with LV ($n = 55$). Using raw scores, the authors reported an overall ES of 4.55 points with significant improvements in three of the four LVQoL subscales—mobility, psychologic, and reading. There was no significant improvement for the activities of daily living subscale. Although raw scores are ordinal, they may not exhibit the essential properties of measurement.[52] Consequently, the

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Table 4. Type of Low Vision Interventions Suggested to the Participants

Low Vision Intervention	Completed, $N = 255$ *n* (%)	Withdrawals, $N = 1016$ *n* (%)	*P*
Refractive correction*	96 (37.6)	777 (76.5)	<0.0001
Low vision devices†			
Telescopes and binoculars for distance vision tasks	30 (11.8)	124 (12.2)	0.85
Magnifiers for near vision tasks	118 (46.3)	437 (43.0)	0.88
Closed circuit television‡	48 (18.8)	239 (23.5)	0.12
Nonoptical devices	91 (35.7)	585 (57.6)	<0.0001
Glare control devices	62 (68.1)	419 (70.3)	
Flashlight for mobility	14 (15.4)	104 (17.5)	
Written communication devices	3 (3.3)	24 (4.1)	
Talking watch	1 (1.1)	2 (0.3)	
White cane	3 (3.3)	11 (1.9)	
Reading lamp	2 (2.2)	9 (1.5)	
Reading stand	1 (1.1)	15 (2.5)	
Needle threader	5 (5.5)	11 (1.9)	
Type of multidisciplinary service accessed by participants			
Optometry led	58 (22.7)	229 (22.5)	
Other	197 (77.3)	787 (77.5)	0.94

DS denotes dioptres.
* Includes single vision glasses and high-add bifocals (up to +4.00 DS).
† One participant could have been advised to use more than one device.
‡ Includes portable video magnifiers and mouse model CCTV.

use of parametric tests on such scores may compromise the analyses and the findings thereof. By comparison, the use of interval-level scores generated through Rasch analysis (as has been done in the present study) legitimizes the use of parametric statistical techniques. In addition, Rasch analysis has been shown to provide dramatically enhanced precision and increased ES compared with traditional psychometric methods.[53] Given this, it would be interesting to see whether the Do et al.[31] findings persist using interval-transformed LVQoL scores.

Following LVR, our participants reported significant improvements in reading, mobility, visual information processing, and visual motor skills assessed using the IVI and VA LV VFQ-48. Given that reading is one of the main areas that LVR is trying to improve, this is a good result, and it is not surprising. Improvement in activities necessary to read bills and read labels or instructions on medicines supports the idea that survival or spot reading activities, which are necessary in the performance of activities of daily living, are improved and should be a focus of LVR services.[12,54] Although both the IVI and VA LV VFQ-48 have been developed and validated primarily in (older) Western populations, the item content was considered relevant by our relatively younger population (except for sports-related activities that were not applicable to approximately 70% of the participants). Nonetheless, Rasch analysis is robust to missing data.[55] Our participants reported large improvements in their reading abilities (ES = 1.0) after LVR as compared to the moderate improvements for mobility (ES = 0.63) and the overall VRQoL (0.63) as assessed using the IVI. By comparison, participants reported moderate improvements for reading (ES = 0.72), mobility (ES = 0.45), visual information processing (ES = 0.54), overall visual ability (ES = 0.67), and visual motor skills (ES = 0.47) as assessed using the

Table 5. Components of Rehabilitation Services Undertaken by the 255 Participants Who Completed the Study

Component	*n* (%)
Safe navigation strategies and training in orientation and mobility*	30 (11.8)
Training in use of computer assistive software	19 (7.5)
Disability certification†	73 (28.6)
Training in activities of daily living‡	5 (1.9)
Employment counseling	32 (12.5)
Educational counseling	36 (14.1)
No rehabilitation services accessed§	58 (22.7)

* Undergoing training in orientation and mobility was suggested to 91 patients.
† A person who has best-corrected visual acuity < 20/200 in better eye or visual field < 20° is eligible for this and can avail him- or herself of government benefits for the disabled in India.
‡ This training was suggested to 35 patients.
§ These participants availed themselves of optometry-led services only.

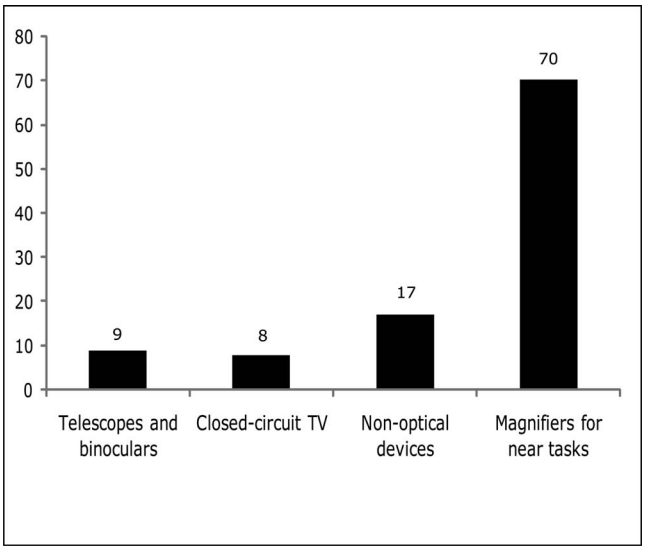


Figure. Bar chart showing the types of low vision devices (*x*-axis) purchased by the participants (*y*-axis; number of participants).

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Table 6. IVI Overall and Subscale Scores (Mean ± 1 SD) at Pre and 4 Months Post Low Vision Rehabilitation (Complete Case Analysis)

IVI Scores	Pre LVR*	Post LVR*	*P* Value	Effect Size (95% CI)
Overall	0.31 ± 1.36	1.17 ± 1.19	<0.0001	0.63 (0.46, 0.80)
Reading and accessing information	0.25 ± 1.65	1.89 ± 2.04	<0.0001	1.0 (0.80, 1.20)
Mobility and independence	0.72 ± 1.82	1.67 ± 2.20	<0.0001	0.53 (0.31, 0.75)
Emotional well-being	0.21 ± 2.46	0.61 ± 2.78	0.06	0.18 (−0.12, 0.48)

* Values in logits.

VA LV VFQ-48. All these results pertain to the 20% of the participants who completed the study ($n = 255$), and these estimates may be biased. While this may reflect an issue with the present study, we believe that it is a larger issue perhaps pointing toward the lack of engagement of patients with VI in the LVR process in general, which is beyond the scope of this paper. We could obtain only a small ES (Tables 8, 9) when we used the BOCF method in our sensitivity analyses from all recruited participants ($n = 1016$) to estimate the ES. Nonetheless, the improvement for both the reading and mobility subscales of the IVI following LVR in our study ($n = 255$) is three to four times larger than that reported by Lamoureux et al.[4] for an older Australian visually impaired population (mean age, 80.3 years; ES = 0.20 for reading and 0.17 for mobility). A possible explanation for this is the difference in the participants' primary cause of LV. In the study by Lamoureux et al.,[4] the proportion of patients who had age-related macular degeneration as the cause of LV (61.9%) was almost 15 times that in our study (4%).[4] The integrity of the central retina is critical for visual performance tasks like reading, face recognition, visual search, and so on.[56,57] Despite optimal magnification, patients with central scotomas usually read more slowly than those with intact central field, and this could have resulted in lower ES for reading in their study.[58] The nature of service provision at VRC, wherein mobility training and other safe navigation strategies were provided by O and M instructors at our center (compared to referrals to community service team), may have resulted in a relatively higher uptake of these services in our study (11.8%) and overall better post-LVR scores for the mobility subscale in our study.

Our results regarding patient-reported improvement ($n = 255$) for the overall IVI, reading, and mobility are also better (almost three to five times larger ES) than those reported recently by Wang et al.[5] for an Australian cohort (mean age, 79.4 years) of patients with LV. They obtained small improvements for the overall IVI score (ES = 0.24) and for mobility (ES = 0.22), but did not find any improvements for reading. Given that their participants had much better pre-LVR scores than ours (ceiling effect) for the overall IVI as well as for the three subscales, one could surmise that that their rehabilitation demand may have been lower. Consequently, the room for improvement following LVR would be expected to be less. In their study, 49.3% of the participants had vision ≥20/60 and/or restricted fields, compared with 32.2% in the present study. Therefore, the better acuity among their participants may have been a possible reason for the much higher pre-LVR scores.

In the Veterans Affairs Low Vision Intervention Trial (LOVIT), Stelmack et al.[19] used the VA LV VFQ-48 in an outpatient setting with older visually impaired participants (mean age, 78.8 years) and demonstrated significant improvements in reading, mobility, visual information processing, and visual motor skills following the LVR program for the treatment group as compared with the control group. Participants in their treatment group reported significantly greater improvements for the reading subscale following LVR as compared to those in our study (2.06 ± 1.2 vs. 1.49 ± 2.15 logits; $P < 0.05$). However, there were no significant differences in improvement overall or for other subscales ($P > 0.05$). A potential explanation for the difference in the improvements in the reading subscale between the two studies may be multiple factors such as differences in VI severity, principal cause of LV, and the provision of optical (stand and pocket magnifiers) and electronic (CCTV) devices at no charge to the participants in the LOVIT study. All these devices render it easy to read small fonts, so it is not surprising that patients in the LOVIT study reported large improvements in the reading subscale. As pointed out by the authors, the coverage of LVDs may affect the acquisition rate of the prescribed LVDs, and this may in turn affect the treatment outcomes. As compared to a 100% acquisition rate for the prescribed LVDs in the Stelmack et al. study, it ranged from 17% (CCTV) to 59% (magnifiers) in our study. In addition, all participants received an extensive amount of standardized training including multiple visits with a LV therapist and homework assignments in that study. In another study that included participants from a blind rehabilitation center and the Visual Impairment Centre to Optimize Remaining Sight, Stelmack et al.[6] showed substantial gains for all the activities when assessed for changes on an item-to-item basis. It may be difficult to directly compare our results with those of Stelmack et al. given that patients were admitted to the hospital for the LVR program for periods ranging from 3 to 42 days. More importantly, the follow-up assessments were carried out immediately prior to discharge from the hospital. Consequently, it is possible that the patients may have overestimated their visual abilities.

As part of our multidisciplinary service, we have an in-house vision-specific counseling service, yet we did not find any significant improvement for the EWB subscale of the IVI after LVR. This lack of improvement in EWB is in accordance with another study suggesting that LVR services may have

Table 7. VA LV VFQ-48 Overall and Subscale Scores (Mean ± 1 SD) at Pre and 4 Months Post Low Vision Rehabilitation (Complete Case Analysis)

VA LV VFQ Scores	Pre LVR*	Post LVR*	*P* Value	Effect Size (95% CI)
Overall	−1.49 ± 1.80	−2.70 ± 2.27	<0.0001	−0.67 (−0.45, −0.89)
Reading	−0.22 ± 2.06	−1.17 ± 2.40	<0.0001	−0.72 (−0.47, −0.97)
Mobility	−1.72 ± 2.45	−2.83 ± 2.65	<0.0001	−0.45 (−0.15, −0.75)
Visual information	−2.45 ± 2.33	−3.72 ± 2.66	<0.0001	−0.54 (−0.25, −0.83)
Visual motor	−2.51 ± 0.14	−3.56 ± 0.15	<0.0001	−0.47 (−0.20, −0.74)

* Values in logits.

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Table 8. Effect of Alternative Analytic Method on the Estimated Magnitude of Treatment Effect on the IVI Overall and Subscale Scores (Mean ± 1 SD) at 4 Months Post Low Vision Rehabilitation

IVI Scores	Pre LVR*	Post LVR*	*P* Value	Effect Size (95% CI)
Overall	0.26 ± 1.72	0.45 ± 1.39	<0.0001	0.16 (0.09, 0.23)
Reading and accessing information	0.60 ± 0.33	0.67 ± 0.42	<0.0001	0.21 (0.19, 0.23)
Mobility and independence	0.56 ± 0.36	0.61 ± 0.42	<0.0001	0.14 (0.12, 0.16)
Emotional well-being	0.76 ± 0.48	0.79 ± 0.50	0.006	0.04 (0.01, 0.07)

* Values in logits.

minimal impact on depressive symptoms but may improve VRQoL outcomes such as vision-specific distress.[59]

Despite a relatively small proportion of participants acquiring the prescribed LVDs, we obtained a nearly large improvement in the overall VF and VRQoL following LVR among those who completed the study. While the multidisciplinary nature of the services rendered may partly explain the result, the fact that the majority of the participants (76.5%) had moderate VI may have also contributed to it. The observed ES is much larger than that obtained for adults in an Australian population.[4,5] Although in line with other studies we did not find any interaction between LVR change and type of service (optometry led or other services), we wish to point out that our study did not have adequate power to detect an interaction of considerable importance (i.e., >0.5).[4,25] Given the tertiary eye care nature of our facility, one would expect most LV patients to have chronic ocular conditions that will require multidisciplinary services, and this was borne out by our data. However, <15% utilized the multidisciplinary services.

A few caveats should be taken into account when one is considering the overall results of our study. These include the high dropout rate (as has been addressed earlier) and a low recruitment rate (34%). We speculate that the lack of engagement with the LVR process is a possible reason for this. A high attrition rate in studies from India is not uncommon and has been reported previously,[60] including the for patients with LV from one of our centers in Eastern India.[61] This latter study also reported low acceptance of LVR services by patients in this region, cited as a major barrier to effective delivery of available services despite easy availability of the services within the institute. The authors further reported that patient perception and awareness are prime factors for successful LVR service delivery and recommended awareness programs for patients regarding the benefits of LVR services.[61] In practical terms, attrition affects the generalizability of results to the wider population by introducing bias, which affects the ability to draw the correct conclusion in the study population and hence the general population. As noted earlier, this aspect needs to be considered in the interpretation of our results, in particular regarding how systematic withdrawals of those from urban areas, those with retinal disorders, and those without systemic comorbidity in the present study threaten to introduce bias in the longitudinal analysis of LVR outcomes.

In the present study, it appears that the withdrawals were not at random ("missing not at random" [MNAR]—that is, the probability of being lost to follow-up depends on the outcome to be measured and cannot be completely explained by covariates[62,63]). However, there is no method to ascertain if the loss to follow-up is MNAR; and in such a scenario, one should assume that loss to follow-up will be MNAR in a cohort study and should plan to provide sufficient resources to achieve the maximum follow-up rate possible.[64] A high rate of loss to follow-up occurred despite attempts to reach our patients over the phone or through letters of reminder. Another important consideration in the interpretation of our results concerns the limited engagement with the multidisciplinary rehabilitation process even among those who completed the study (Table 5). The lack of a residential facility at our center to enable provision of various types of training (activities of daily living, orientation and mobility, and so on) may have affected the utilization rate of these services given that these involve extended sessions requiring patients to visit the center frequently. It is also plausible that participants did not feel the need ("felt need") given the low acceptance of these services.[61] Taken together, the results of our study demonstrate the importance of considering the effect of incomplete follow-up on data acquired in LVR outcome studies.

Our study did not include a control group for comparison because we felt that it would be unethical to withhold any type of LVR service from patients. Thus it is difficult to comment on the possibility of worsening of the overall visual disability and VRQoL in the absence of any LVR. Nonetheless, it may be possible and easier to randomly assign participants to one of two types of LVR services (optometry-led services and other multidisciplinary services), and this should be considered in future studies.

In conclusion, our study demonstrates large improvements in VF and VRQoL in adults with LV through provision of LVR services at VRC, LVPEI in South India. The results also demonstrate the importance of considering the effect of high attrition on data acquired in longitudinal studies of outcomes of LVR. Bias toward more favorable outcomes may occur when dropout rates (withdrawals) are high (as has been the case with our study), so extrapolation of findings from this study may be limited due to sample characteristics. However, this study has provided a foundation for a large-scale longitudinal investigation in this field. Patient education and the building of

Table 9. Effect of Alternative Analytic Method on the Estimated Magnitude of Treatment Effect on the VA LV VFQ-48 Overall and Subscale Scores (Mean ± 1 SD) at 4 Months Post Low Vision Rehabilitation

VA LV VFQ scores	Pre LVR*	Post LVR*	*P* Value	Effect Size (95% CI)
Overall	0.30 ± 0.14	0.33 ± 0.22	<0.0001	0.21 (0.20, 0.22)
Reading	0.66 ± 0.42	0.69 ± 0.45	0.001	0.07 (0.05, 0.09)
Mobility	−1.78 ± 2.44	−2.03 ± 2.54	<0.0001	−0.10 (−0.04, −0.24)
Visual information	−2.20 ± 2.09	−2.47 ± 2.25	<0.0001	−0.13 (−0.01, −0.25)
Visual motor	−2.36 ± 2.10	−2.59 ± 2.20	<0.0001	−0.11 (−0.01, −0.23)

* Values in logits.

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awareness programs about the potential benefits of LVR, including providing details on specific types of patients who are likely to benefit from these services, may help with the recruitment process for longitudinal studies. Although many activities were improved, EWB was much less likely to improve. One suggestion is that LVR programs might target mental health issues more explicitly. However, this also implies that specific LVR efforts targeted toward improvement in mental health/EWB of patients are reimbursed and that the persons providing these services have appropriate training. Low vision rehabilitation proved relatively effective at improving the functions it targeted. These results should encourage eye care professionals (both ophthalmologists and optometrists) to refer adults with LV for LVR services. Further research is required to understand if the ES obtained at 4 months post LVR can be sustained over a longer period and also to determine specifically which of the components of the LVR services are used maximally by adults with LV in order to effectively plan increases in these services.

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